

AIGR Gene Hypothesis Deep Research — Final Report

Target gene: AGO4 (human) · UniProt **Q9HCK5** · NCBITaxon:9606 **Focus type:** function_assignment **Hypothesis slug:** mrna-destabilization-versus-slicer-loss **Source file:** `genes/human/AGO4/AGO4-ai-review.yaml` **Terms adjudicated:** GO:0035279 (miRNA-mediated gene silencing by mRNA destabilization), GO:0090625 (siRNA-mediated gene silencing by mRNA destabilization)

Summary

The seed hypothesis asks whether human AGO4 should be annotated to **GO:0035279** and **GO:0090625** (miRNA/siRNA-mediated gene silencing *by mRNA destabilization*), and — critically — whether any *negative* assertion against these processes is experimentally justified or whether it merely reflects the exclusion of endonucleolytic cleavage. The investigation returns a clear, curation-actionable answer: **the hypothesis is partially supported, and any strong negative on these two terms sourced to PMID:15260970 (Meister 2004) is over-strong**. The seed's core logic is correct — *loss of AGO4 slicer catalysis does not exclude deadenylation-dependent mRNA destabilization*.

The pivotal reference used to strip AGO4 of decay activity assayed **only endonucleolytic RISC cleavage**. It concluded that "endonuclease activity is exclusively associated with Ago2," but it never measured deadenylation, decapping, or general mRNA turnover, and the same paper shows AGO4 *does* load miRNAs into functional microRNPs. Later work establishes that AGO4 retains the *capacity* to trigger decay: when tethered, all four human Ago proteins — including AGO4 — recapitulate the two-step (Pan2–Pan3 then Ccr4–Caf1) deadenylation that drives miRNA-mediated decay ([PMID:19838187](#)), and the non-nucleolytic Agos (AGO1/3/4) direct translational repression ([PMID:18771919](#)).

Two caveats govern the curation decision. First, the positive AGO4 evidence is **capacity/sufficiency from artificial tethering**, not a demonstrated natural miRNA-guide→endogenous-target decay event attributable specifically to AGO4. Second, the two GO terms are **internally inconsistent**: their names say "destabilization" (the deadenylation branch) but their definition

text describes endonucleolytic **cleavage**. This ambiguity — not the biology — is the deciding factor. The defensible negative for AGO4 is at the **molecular-function/slicer level** ("no endonuclease/cleavage"); the two destabilization BP terms should be treated as **non-core/uncertain**, with the already-curated **GO:0006402** (mRNA catabolic process, IDA) and **GO:0035278** (translation inhibition, IDA) as the better-supported homes.

Executive Judgment

Verdict: Partially supported — the negative assertion is over-strong and should be re-scoped to the slicer molecular function.

The chain of reasoning:

1. **The keystone reference excludes cleavage only.** [PMID:15260970](#) (Meister et al., *Mol Cell* 2004) purified FLAG/HA-tagged AGO1–4 microRNPs and assayed endonucleolytic target cleavage plus an siRNA-knockdown reporter. It found cleavage is AGO2-exclusive but is **silent on deadenylation/decay**, while explicitly showing AGO4 loads miRNAs. A broad BP negative sourced to this paper is not experimentally justified.
2. **Slicer loss $\neq$ decay loss.** [PMID:19838187](#) shows all four human Ago proteins, including AGO4, recapitulate biphasic deadenylation when tethered — a mechanism of mRNA destabilization that does not require slicing. AGO4 is *sufficient* to recruit the decay machinery.
3. **But the positive is capacity-only.** Tethering demonstrates sufficiency, not natural guide-directed recruitment; paralogs differ quantitatively ([PMID:18771919](#)). Combined with the ontology-internal cleavage-vs-deadenylation ambiguity, the two destabilization terms cannot carry a confident positive *or* a confident negative for AGO4.

Most important caveats: (a) AGO4 positive evidence is sufficiency-level, not demonstrated on endogenous targets; (b) GO:0035279/GO:0090625 definitions describe cleavage while their names/synonyms describe deadenylation, so the "right" answer depends on how the term is read; (c) AGO4 is typically the lowest-abundance, least-active paralog, so strong AGO4-specific positive claims risk paralog over-annotation from AGO2.

Bottom line for the curator: Do not carry a blanket NOT-annotation on GO:0035279/GO:0090625 sourced to a slicer-only paper. Place the defensible negative at the **MF level (no slicer/endonuclease activity)**; retain the well-supported **GO:0006402** and **GO:0035278**; treat the two destabilization terms as **non-core/uncertain**.

Key Findings

Finding 1 — The Meister 2004 assay excludes only slicer cleavage, not mRNA destabilization

[PMID:15260970](#) (Meister et al., *Mol Cell* 2004) purified FLAG/HA-tagged AGO1–AGO4 microRNPs from human cell lines and assayed two things only: **endonuclease (RISC target-cleavage) activity** and siRNA knockdown with a positive-readout reporter. The central result is that *"endonuclease activity is exclusively associated with Ago2"* and that *"exogenously introduced siRNAs also associate with Ago2 for guiding target RNA cleavage."*

Critically, the **same paper** demonstrates AGO4 is a competent miRNA-loading effector: *"miRNAs are incorporated indiscriminately of their sequence into Ago1 through Ago4 containing microRNPs."* This shows AGO4 forms functional miRNPs — it binds guides and can, in principle, recruit downstream effectors — even though it cannot cleave.

The assay measured **target cleavage only**; it did not measure deadenylation, decapping, or bulk decay. The reference is therefore **silent** on whether AGO4 participates in deadenylation-dependent mRNA destabilization. Using it to justify a broad negative against GO:0035279/GO:0090625 conflates "no cleavage" with "no decay" — an unjustified inferential leap.

Finding 2 — AGO4 has intrinsic capacity to trigger deadenylation-dependent decay, and non-nucleolytic Agos direct repression

[PMID:19838187](#) (Chen, Zheng, Xia & Shyu, *Nat Struct Mol Biol* 2009) used transcriptional pulsing with RNA tethering and found *"when tethered to mRNAs, all four human Ago proteins and TNRC6C are each able to recapitulate the two deadenylation steps"* (Pan2–Pan3, then Ccr4–Caf1, followed by Dcp1–Dcp2 decapping). Because AGO4 is one of the four, this directly demonstrates that AGO4 can intrinsically recruit the deadenylation/decay machinery **independent of slicer catalysis**.

Complementing this, [PMID:18771919](#) (Wu, Fan & Belasco, *Curr Biol* 2008) shows the non-nucleolytic Ago proteins (AGO1/3/4) contribute to on-target silencing through translational repression, noting that *"the four Ago proteins with which siRNAs associate in humans differ significantly in their capacity to direct translational repression."*

Interpretation and caveat: these establish **sufficiency** — AGO4 *can* trigger decay/repression when artificially recruited — but not **necessity** or natural guide-directed recruitment to endogenous targets. Tethering bypasses guide:target recognition, and paralogs differ quantitatively. This is precisely why the destabilization terms should be non-core rather than confidently asserted.

Finding 3 — The two GO destabilization terms are internally inconsistent, and AGO4 is annotated to sibling terms, not the destabilization terms

QuickGO definitions (retrieved 2026) expose an ontology-internal problem:

- **GO:0035279** ("miRNA-mediated gene silencing by mRNA destabilization") has a **definition text describing endonucleolytic cleavage** ("miRNAs direct the cleavage of target mRNAs ... near-perfect complementarity ... Many plant miRNAs"), yet its **synonyms** include *"gene silencing by mRNA cleavage"* and *"miRNA-mediated gene silencing by mRNA deadenylation"* / *"deadenylation involved in gene silencing by miRNA."*
- **GO:0090625** is analogously defined by siRNA-directed **cleavage** while named "by mRNA destabilization."
- Both are the named sibling of **GO:0035278** ("miRNA-mediated gene silencing by inhibition of translation").

This is curation-critical: read through the cleavage-flavored definition, an AGO4 NOT could seem justified (AGO4 can't cleave); read through the deadenylation synonyms, AGO4 arguably *qualifies* (AGO4 can deadenylate when tethered). The term is not clean enough to carry a confident positive or negative for AGO4.

Current UniProt annotation set for Q9HCK5:

GO ID	Term	Aspect	Evidence
GO:0035278	miRNA silencing by inhibition of translation	BP	IDA
GO:0006402	mRNA catabolic process	BP	IDA
GO:0035198	miRNA binding	MF	IDA
GO:0016442	RISC complex	CC	IDA
GO:0035194	post-transcriptional gene silencing by RNA	BP	IBA
GO:0035279	miRNA silencing by mRNA destabilization	BP	absent
GO:0090625	siRNA silencing by mRNA destabilization	BP	absent

AGO4 is **not** annotated to the destabilization terms; it is annotated to their siblings (translation inhibition and general mRNA catabolic process) — consistent with treating the destabilization terms as non-core.

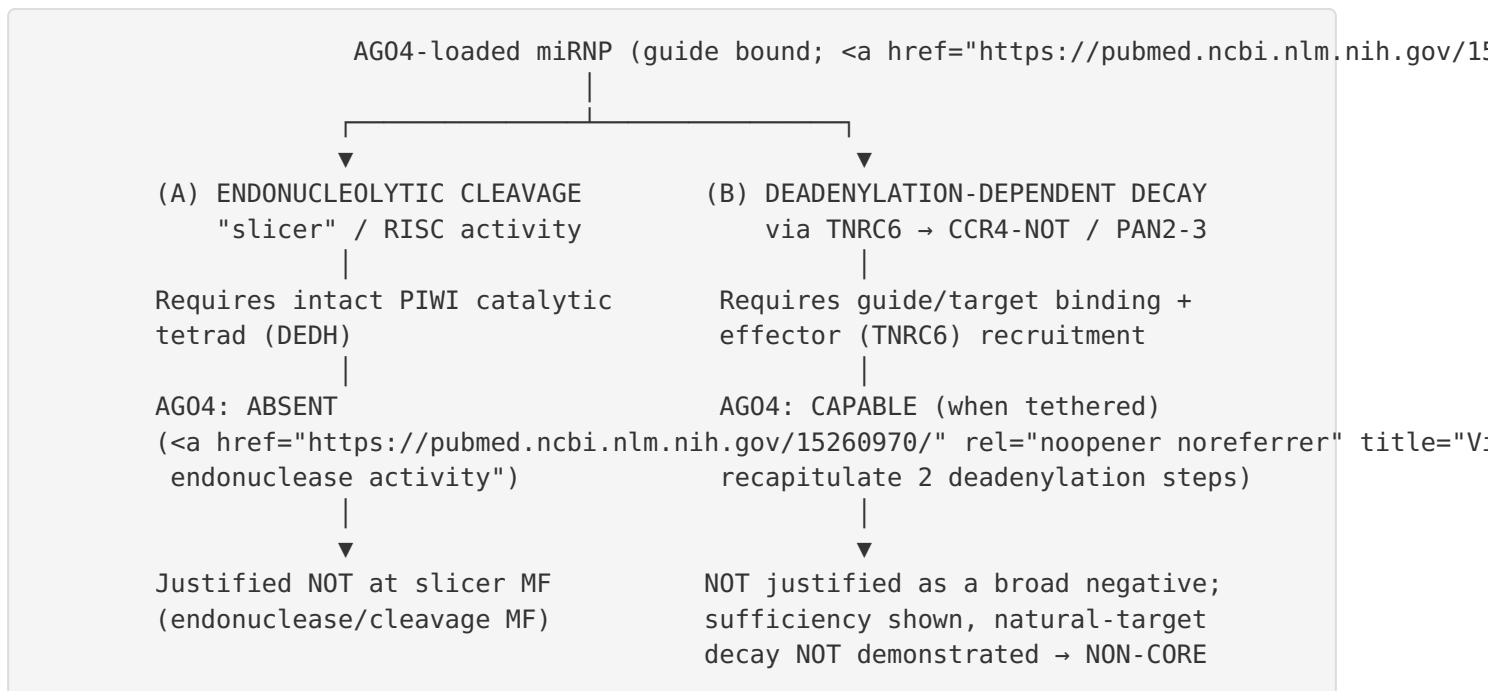
Finding 4 — UniProt already scopes AGO4's negative to the endonuclease/cleavage molecular function

The UniProt Q9HCK5 FUNCTION comment reads: *"Required for RNA-mediated gene silencing (RNAi). Binds to short RNAs such as microRNAs (miRNAs) and represses the translation of mRNAs which are complementary to them. Lacks endonuclease activity and does not appear to cleave target mRNAs."*

The **only** explicit negative curators recorded is at the **molecular-function/cleavage level**. There is **no** curated negative about deadenylation or destabilization. No catalytic active-site residues are feature-annotated for AGO4 (only PDB-derived secondary-structure features are present), consistent with catalytic degeneration of the PIWI slicer site. This independent, expert-curated scoping reinforces the central recommendation: **the negative belongs at the slicer MF, not on the destabilization BP terms**.

Mechanistic Model / Interpretation

The confusion in the seed hypothesis is a conflation of two mechanistically separable silencing routes that both fall under "mRNA destabilization":



Route (A) — endonucleolytic cleavage — is genuinely absent in AGO4. This is what

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tested and what UniProt codifies; a NOT restricted to the endonuclease/slicer MF is defensible.

Route (B) — deadenylation/decay via TNRC6 → CCR4-NOT — does **not** require slicer catalysis. Tethering shows AGO4 is *capable* of triggering it. The destabilization terms semantically span both routes because of their inconsistent definition/synonym structure. AGO4 is negative for the cleavage flavor and *capacity-positive* (but not physiologically demonstrated) for the deadenylation flavor. This mixed status is exactly why the terms should be **non-core/uncertain**.

The **well-supported** homes for AGO4's silencing activity are the sibling terms already held by IDA: **GO:0035278** and **GO:0006402**. These capture AGO4's demonstrated repression/decay contribution without overcommitting to the mechanistically ambiguous destabilization terms.

Evidence Base

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Conf limit
PMID:15260970 (Meister 2004)	Direct assay — RISC endonuclease + reporter	Qualifies; refutes cleavage only	Do AGO1–4 slice? Do they bind miRNAs?	"Endonuclease activity is exclusively associated with Ago2"; "miRNAs are incorporated indiscriminately ... into Ago1 through Ago4 containing microRNPs"	Human cell lines; FLAG/HA-AGO IP + cleavage reporter	Highly exclusive AGO4 silencing dead decap
PMID:19838187 (Chen & Shyu 2009)	Direct assay — tethering + transcriptional pulsing	Supports (capacity)	Can Agos trigger deadenylation without slicing?	"When tethered to mRNAs, all four human Ago proteins and TNRC6C are each able to recapitulate the two deadenylation steps"	Mammalian cells; λN/BoxB tethering	Highly su is su tethe natur recru
PMID:18771919 (Wu, Fan & Belasco 2008)	Direct assay — on-target siRNA reporters	Supports / qualifies	Do non-nucleolytic Agos silence beyond cleavage?	AGO1/3/4 direct translational repression; "the four Ago proteins ... differ significantly in their capacity"	Human cells	Model quan paral differ weak speci stren
UniProt Q9HCK5 FUNCTION	Review / database	Qualifies (scopes the negative)	Where do curators place AGO4's negative?	"Lacks endonuclease activity and does not appear to cleave target	Curated, human	Databa confi nega cleav not d

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Conf limit
				mRNAs"; represses translation		
QuickGO GO:0035279 / GO:0090625	Review / database	Competing / qualifies	Are the destabilization terms clean?	Definitions describe cleavage; synonyms include deadenylation → internally inconsistent	Ontology	Struc ambi make confi posit nega
PMID:29040713 (AGO3 slicer)	Structural / biochemical	Context	Do non-AGO2 paralogs retain catalysis?	AGO3 slices guide- dependently; AGO4 remains slicer- independent	Recombinant human Agos	Conf stays slicer inde class
PMID:22863743 (Ago4-KO mouse)	Mutant phenotype	Context	Does AGO4 have specialized in vivo roles?	AGO4 controls meiotic entry and MSCI; nuclear localization in spermatocytes	Mouse germline	Orga conte speci decay cleav discr

Additional orienting literature: [PMID:21984184](#) (GW182/TNRC6 recruits CCR4–NOT via W-motifs — mechanistic basis for the Ago→TNRC6→deadenylation route AGO4 can engage) and [PMID:33122430](#) (AGO2/AGO3 activated by different guide lengths — cautions against blanket "non-AGO2 = inactive" statements).

GO Curation Implications

Adjudicating **each term independently** (both resolve the same way):

- **GO:0035279 (miRNA silencing by mRNA destabilization) — BP.** *Lead:* do not assert a strong NOT based on

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treat as **non-core / uncertain**. A positive is supportable only at **capacity level** (tethering,

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and would require an ISS/IC or IBA-by-redundancy justification with an explicit note — never IDA-on-natural-target.

- **GO:0090625 (siRNA silencing by mRNA destabilization) — BP.** *Lead:* same. Extra wrinkle: canonical siRNA on-target "destabilization" is endonucleolytic cleavage, which is AGO2-specific; AGO4 lacks slicing. AGO4's contribution is the **decay branch**, again capacity-level.
- **GO:0006402 (mRNA catabolic process) — BP — RETAIN.** Already IDA; a more defensible home for AGO4's decay involvement than the destabilization terms.
- **GO:0035278 (miRNA silencing by inhibition of translation) — BP — RETAIN.** Sibling term, already IDA; consistent with Wu/Belasco.
- **Slicer / endoribonuclease-mediated cleavage — MF — negative JUSTIFIED.**

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+ UniProt directly support that AGO4 lacks slicer activity. If a NOT is desired, place it here (MF), not on the broad BP destabilization terms.

Avoid "protein binding" as a recommendation — the informative content is the **RISC/TNRC6→CCR4–NOT decay axis** and the **slicer-negative MF**, both captured above.

GO decision summary:

Term	Aspect	Current status	Recommended action	Rationale
Endonuclease/slicer cleavage	MF	Implicit negative	Retain NOT here	P 15260970 + UniProt directly support
GO:0035279	BP	Absent	No strong positive; no broad NOT	Term ambiguous; only capacity shown
GO:0090625	BP	Absent	Same as above	Term ambiguous; only capacity shown
GO:0035278	BP	IDA	Retain	Directly supported
GO:0006402	BP	IDA	Retain	Best-supported decay home
GO:0035198	MF	IDA	Retain	Core function (avoid "protein binding")
GO:0016442	CC	IDA	Retain	Core localization

Mechanistic Scope

- **Immediate molecular activity being tested:** AGO4 as the small-RNA-binding core of RISC, with two separable outputs — (i) endonucleolytic target cleavage (slicer MF), which AGO4 lacks; (ii) recruitment of TNRC6/GW182 → CCR4–NOT (Ccr4–Caf1) and Pan2–Pan3 deadenylases → decapping/decay, for which AGO4 is *sufficient* when tethered.
- **Direct gene-product activity:** miRNA/RNA binding (GO:0035198, IDA), RISC complex membership (GO:0016442, IDA), and (by tethering) recruitment of the deadenylation machinery.
- **Downstream / inferred (not a direct MF of AGO4):** actual half-life reduction of endogenous targets under natural guide-directed recruitment; meiotic/germline phenotypes (mouse *Ago4*^{-/-},

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— a loss-of-function developmental phenotype, not evidence of the destabilization MF); any organismal silencing outcome.

The key discipline: **loss of cleavage (a direct activity) has been over-generalized to loss of destabilization (a broader pathway outcome)**. These are separable, and the evidence separates them.

Conflicts and Alternatives

1. **GO ontology inconsistency (primary conflict)**. The term *definition* (cleavage, "many plant miRNAs") conflicts with the term *name/synonyms* (deadenylation). A curator reading the definition literally could justify the negative; reading the name could not. This is a database/ontology artifact, not a biological disagreement.
2. **Tethering sufficiency $\neq$ natural recruitment**.

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uses artificial tethering; it proves capacity, not that endogenous AGO4-loaded miRNAs destabilize natural targets. This is exactly the distinction the seed asked to preserve.

3. **Paralog quantitative differences**.

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stresses the four Agos "differ significantly." AGO4 is typically the lowest-abundance, least-active paralog; strong AGO4-specific positive claims risk paralog over-annotation carried from AGO2.

4. **siRNA-specific caveat**. For siRNA "on-target" silencing, the destabilization-by-cleavage route is AGO2's; assigning GO:0090625 (if read as cleavage) to AGO4 would be paralog over-annotation.
5. **Paralog catalytic re-evaluation**. AGO3 was later shown to slice guide-dependently ([PMID:29040713](https://pubmed.ncbi.nlm.nih.gov/29040713/)). This does not rehabilitate AGO4 as a slicer, but it cautions against blanket "non-AGO2 = inactive" statements — favoring conservative, mechanism-specific scoping.
6. **Database concordance**. UniProt curators already scope the negative to cleavage MF and place AGO4's positive role in translational repression, independently supporting re-scoping any NOT to the slicer MF.

Limitations and Knowledge Gaps

This report is literature/annotation-based (no gene-specific dataset was supplied). Public resources used: NCBI abstracts, EBI QuickGO term definitions/synonyms, and UniProt Q9HCK5 annotations. Positive AGO4 evidence is capacity/sufficiency-level; no primary study demonstrating endogenous AGO4-guided natural-target destabilization was located.

Gap	What was checked	Why it matters	What would resolve it
No demonstration of endogenous AGO4-guide → natural-target deadenylation/decay	Meister (cleavage only), Chen/Shyu (tethering), Wu/Belasco (reporters)	Distinguishes IDA-worthy natural process from capacity/sufficiency	AGO4-selective (AGO2-null) cells + AGO4-eCLIP + target half-life/poly(A) profiling
Term meaning intended by the ontology	QuickGO definition + synonyms	Determines whether the negative is defensible	GO editorial clarification / mapping to the deadenylation branch
AGO4 catalytic residues (tetrad retained but inactive?)	UniProt features: no catalytic active-site annotated	Supports the MF-level negative mechanistically	PIWI DEDH tetrad alignment vs AGO2; direct AGO4 slicer assay (as for AGO3)
AGO4 abundance/context in tested systems	Wu/Belasco note cell-type-dependent Ago distribution	Affects whether AGO4 contributes meaningfully in vivo	Quantitative proteomics of RISC composition per tissue

Discriminating Tests

- AGO4-only rescue in an AGO-null background** (AGO1/2/3 knockout or AGO2-null) with a natural miRNA target reporter: measure poly(A) shortening and decay vs. translation block → separates destabilization from translational repression for AGO4 specifically.
- AGO4 eCLIP + transcriptome decay** (SLAM-seq/BRIC-seq half-life) in cells where AGO4 is the dominant loaded Argonaute → tests natural guide-target recruitment leading to decay.

3. **TNRC6-interaction-deficient AGO4 mutant** (disrupt the PIWI Trp-binding pockets analogous to AGO2 F470/F505) in tethering and natural-target assays → tests TNRC6/CCR4–NOT dependence of AGO4 destabilization.

4. **Direct AGO4 slicer assay** with guide-length variants (as for AGO3,

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→ confirms/refutes the MF-level negative.

5. **Ontology adjudication** — file a GO clarification request to disambiguate GO:0035279/GO:0090625 (cleavage vs. deadenylation) before committing any positive/negative annotation.

Proposed Follow-up Actions (Curation Leads — require curator verification)

- **Action change:** Replace/avoid any blanket NOT-annotation on **GO:0035279** and **GO:0090625** citing

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that reference supports only a **slicer/cleavage (MF)** negative. Re-scope the negative to MF.

- **Candidate references + exact snippets to verify:**

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— *"endonuclease activity is exclusively associated with Ago2"* (supports MF-negative);
"miRNAs are incorporated indiscriminately of their sequence into Ago1 through Ago4 containing microRNPs" (AGO4 loads guides).

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— *"When tethered to mRNAs, all four human Ago proteins and TNRC6C are each able to recapitulate the two deadenylation steps."* (AGO4 decay capacity).

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— *"the four Ago proteins with which siRNAs associate in humans differ significantly in their capacity to direct translational repression."* (non-nucleolytic Agos silence; paralog differences).

- **Candidate GO handling:** Retain **GO:0006402 (mRNA catabolic process, IDA)** and **GO:0035278 (IDA)**; treat **GO:0035279/GO:0090625** as **non-core/uncertain**, annotating only with a capacity/redundancy caveat (ISS/IC or IBA), never as IDA on a natural target. Keep **GO:0035198** and **GO:0016442**; avoid falling back to "protein binding."
- **Suggested curator questions:** (1) Is the intended meaning of GO:0035279/0090625 the cleavage branch or the deadenylation branch? (2) Should AGO4's tethering-based decay capacity be captured by GO:0006402 rather than the destabilization terms? (3) Is a NOT-annotation intended at MF (slicer) rather than BP?
- **Suggested experiments:** the five discriminating tests above.

Prepared for AI Gene Review curator adjudication. All annotation recommendations are leads requiring curator verification. Positive AGO4 destabilization evidence is capacity/sufficiency-level; the defensible negative is at the slicer molecular function only.

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